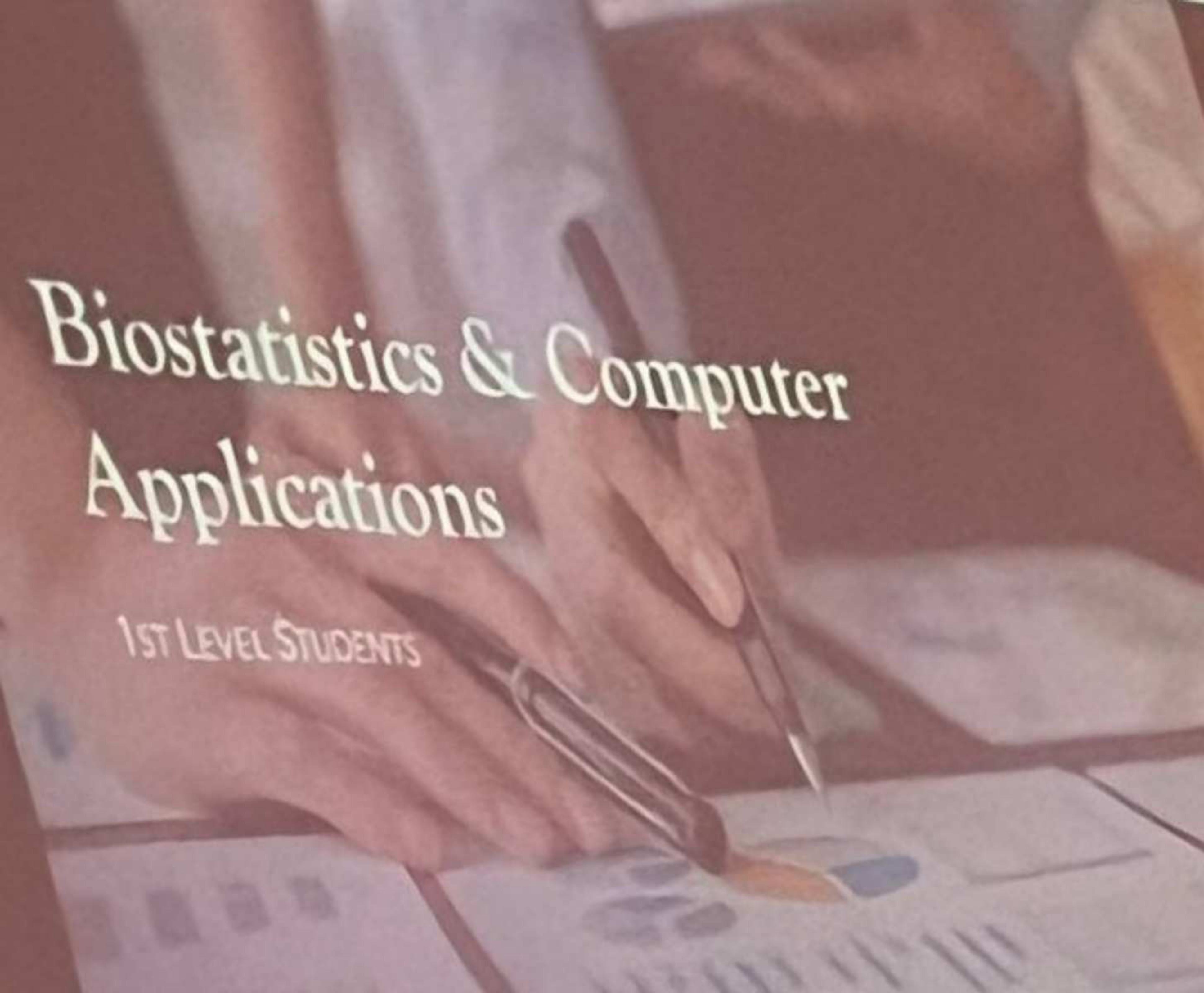




Biostatistics & Computer Applications

1ST LEVEL STUDENTS

Social Issues

مقرر
فضايا مجتمعية
Social Issues

حضور 4 أنشطة

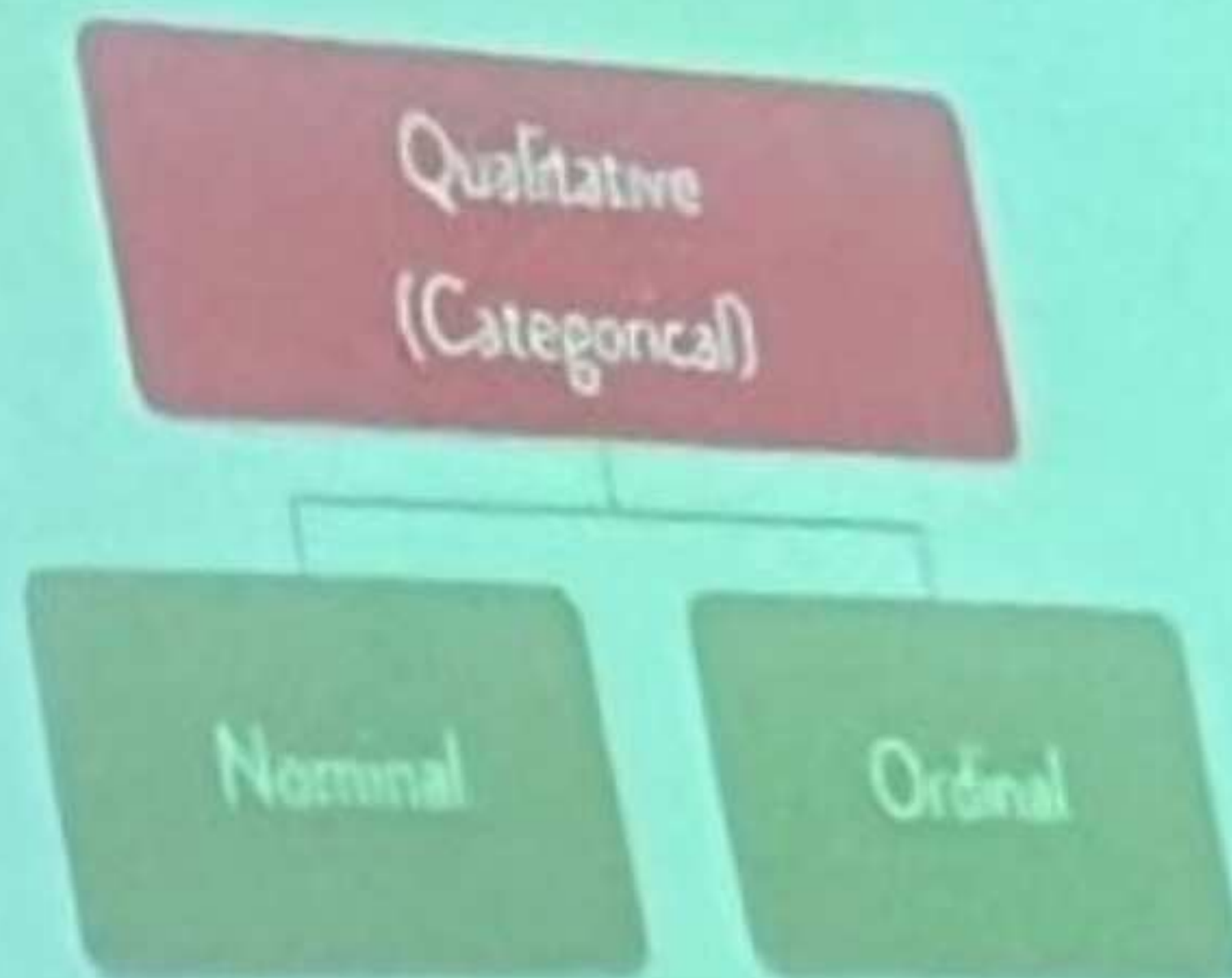
50% (مطلوب نجاح)

امتحان نظري

(50%)

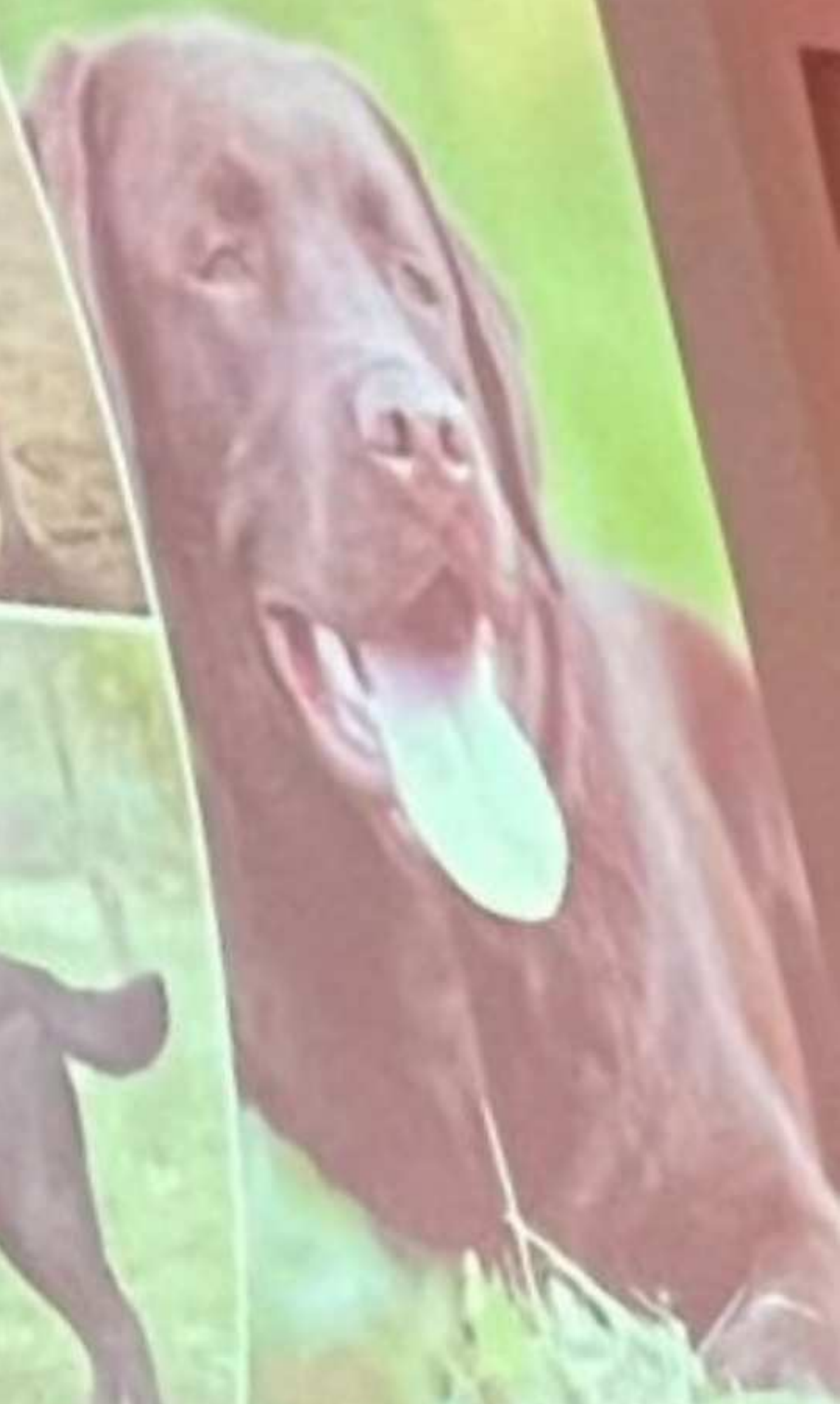
Qualitative (Categorical) Variables

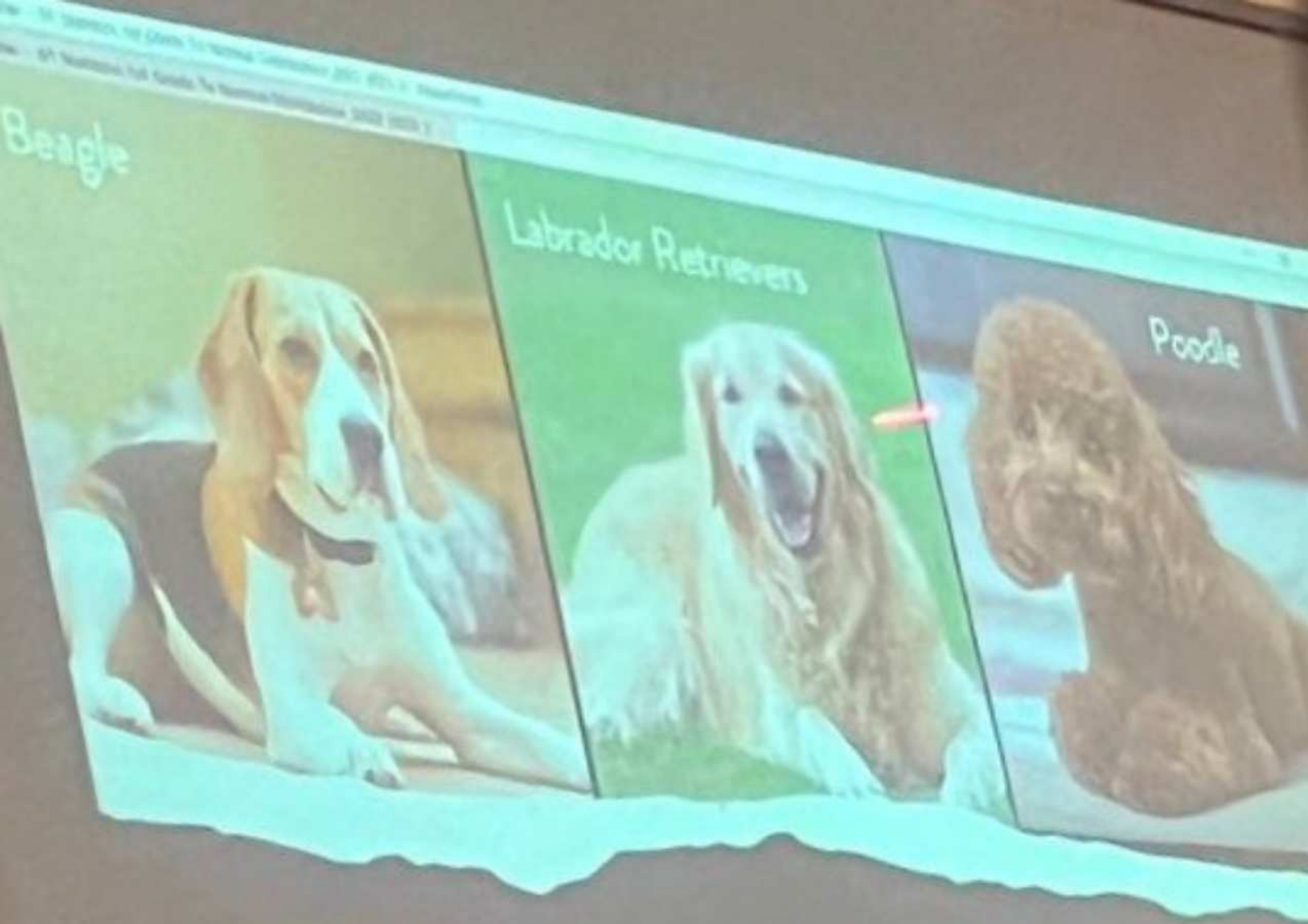
- Qualitative variables describe non-numeric characteristics.



Coat color of
Labrador Retriever
dogs

Yellow, brown, black





Breed of a dog



Gender



Qualitative: Nominal Variables

Nominal Variables:

- These are categories without a specific order.

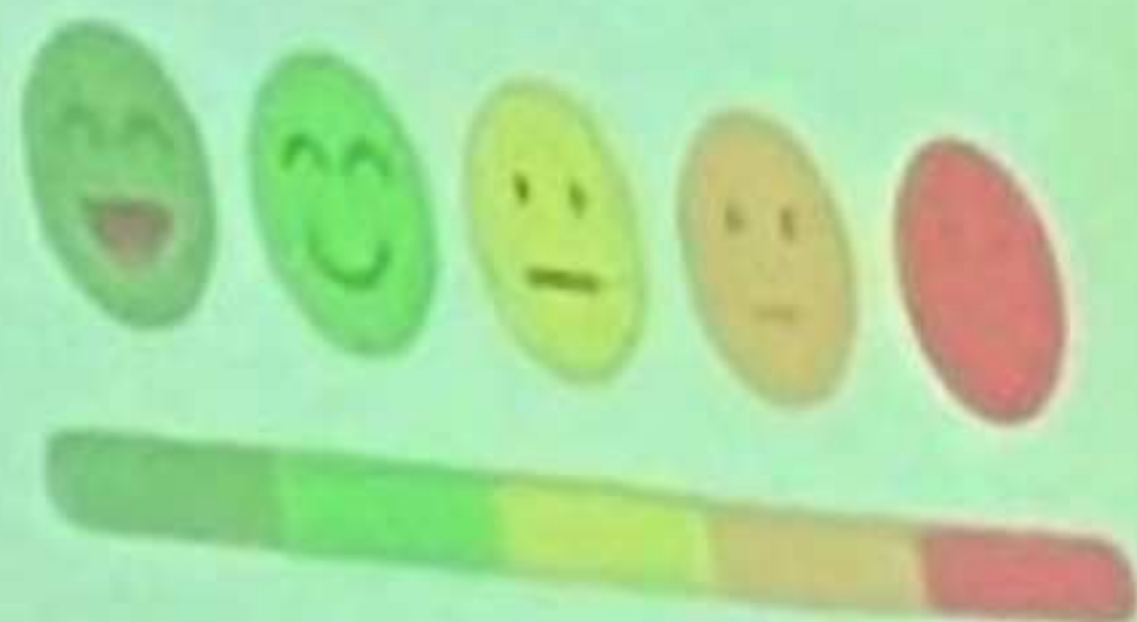
Qualitative: Ordinal Variables

Ordinal Variables:

- They have specific order but no consistent difference between values.

Satisfaction level

1. Very satisfied
2. Satisfied
3. Neutral
4. Unsatisfied
5. Very unsatisfied



Qualitative: Ordinal Variables

Stage of cancer (0 to IV)

Stage 0:

Cancer in situ
(early, no spread)

Stage I:

Small, localized
cancer

Stage II:

Larger or deeper
tumor, may involve
nearby lymph nodes

Stage III:

Larger spread,
regional lymph
nodes involved

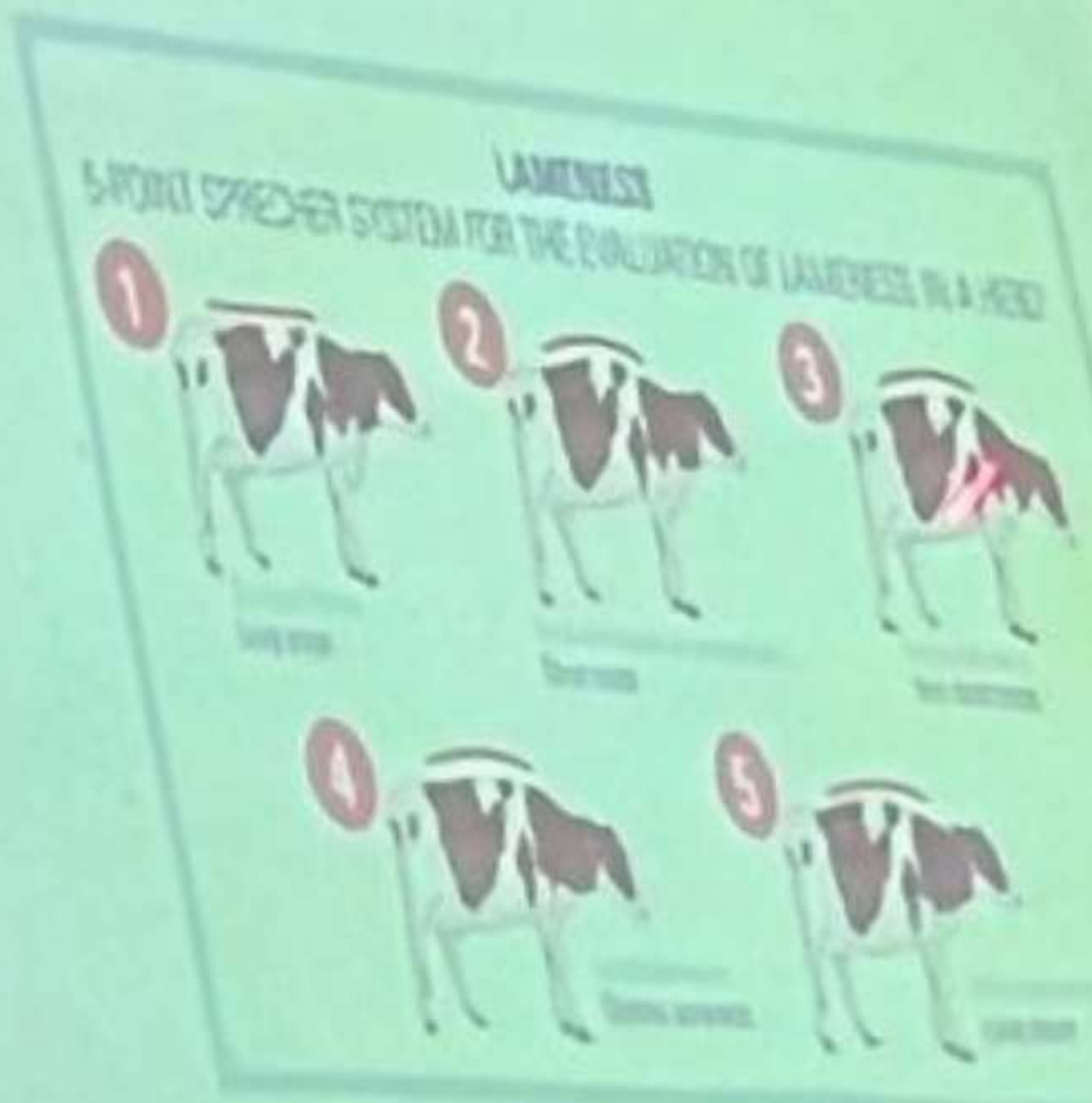
Stage IV:

Cancer has
spread to distant
parts of the body

Qualitative: Ordinal Variables

The pain level in a lame cow

1. Normal
2. Mildly lame
3. Moderately lame
4. Lame
5. Severely Lame



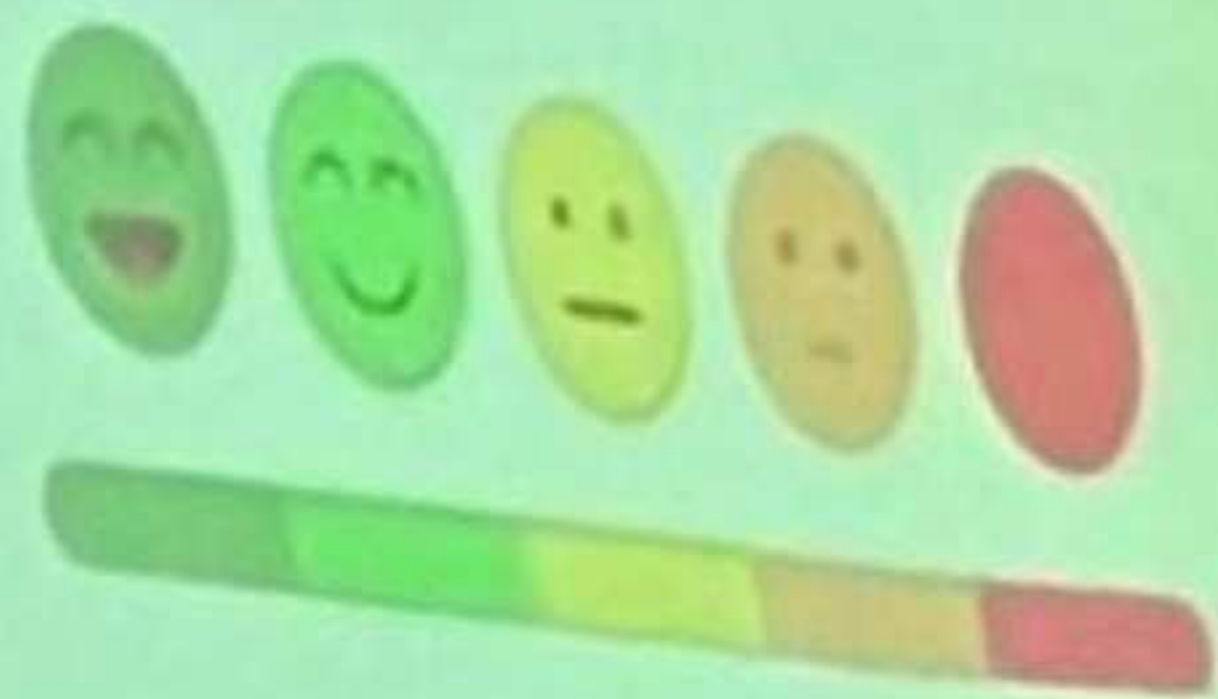
Qualitative: Ordinal Variables

Ordinal Variables:

- They have specific order but no consistent difference between values.

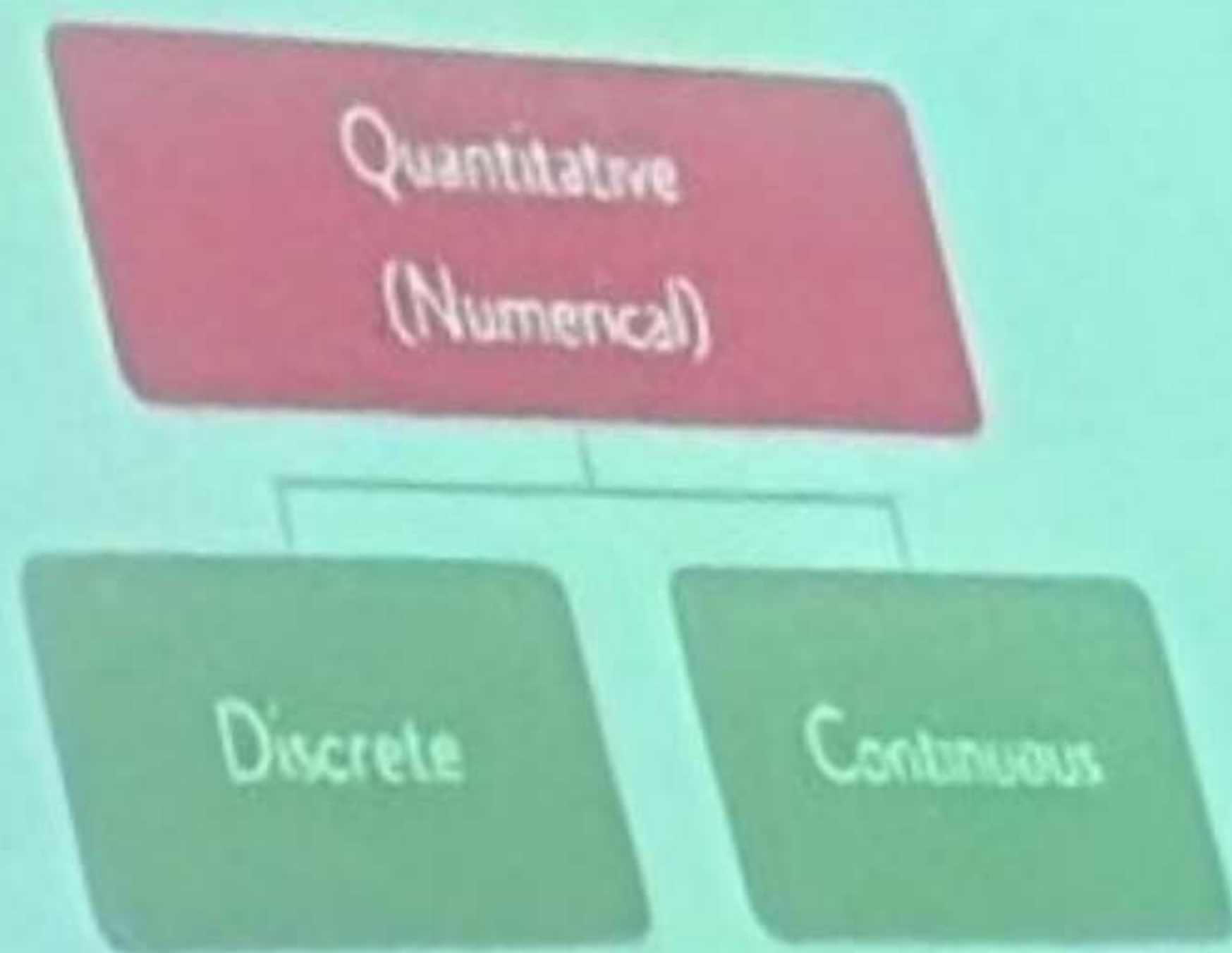
Satisfaction level

1. Very satisfied
2. Satisfied
3. Neutral
4. Unsatisfied
5. Very unsatisfied



Quantitative (Numerical) Variables

- Quantitative variables are numeric and can be measured.



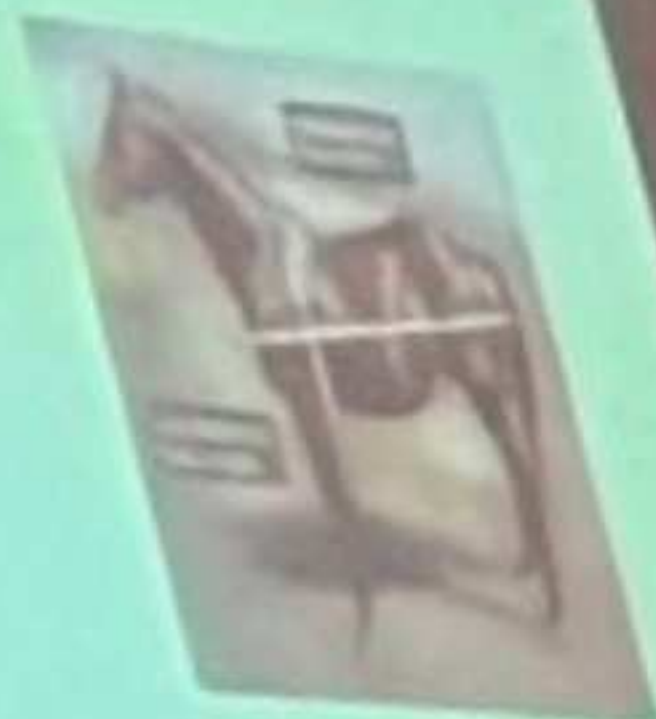
Quantitative: Continuous Variables

Continuous Variables:

- These can take any value within a range and are measurable.

Examples

- Weight of a horse (e.g., 450 kg, 455.5 kg).
- Blood glucose level
- Milk production
- Speed



NB. In SPSS, both discrete and continuous variables are termed "Scale variables"

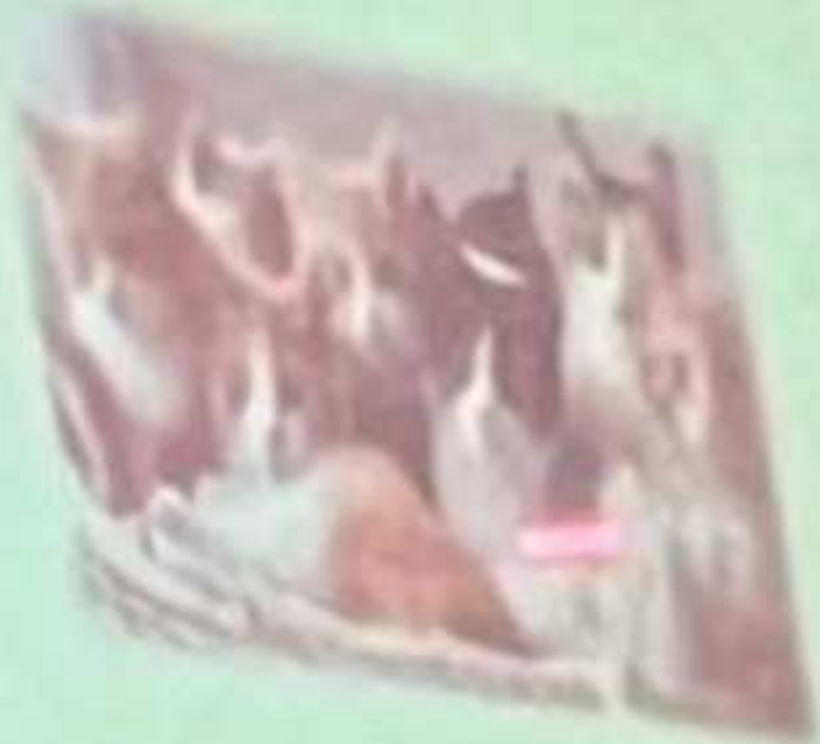
Quantitative: Discrete Variables

Discrete Variables

- These are countable values (usually integer values).

Examples

- Number of vaccinations a puppy has received (e.g., 1, 2, 3).
- Litter size
- Number of pregnancies
- Number of egg



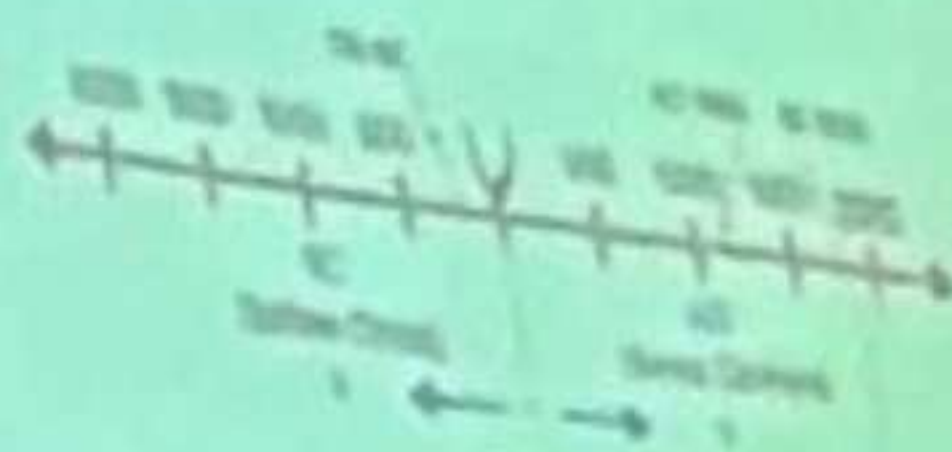
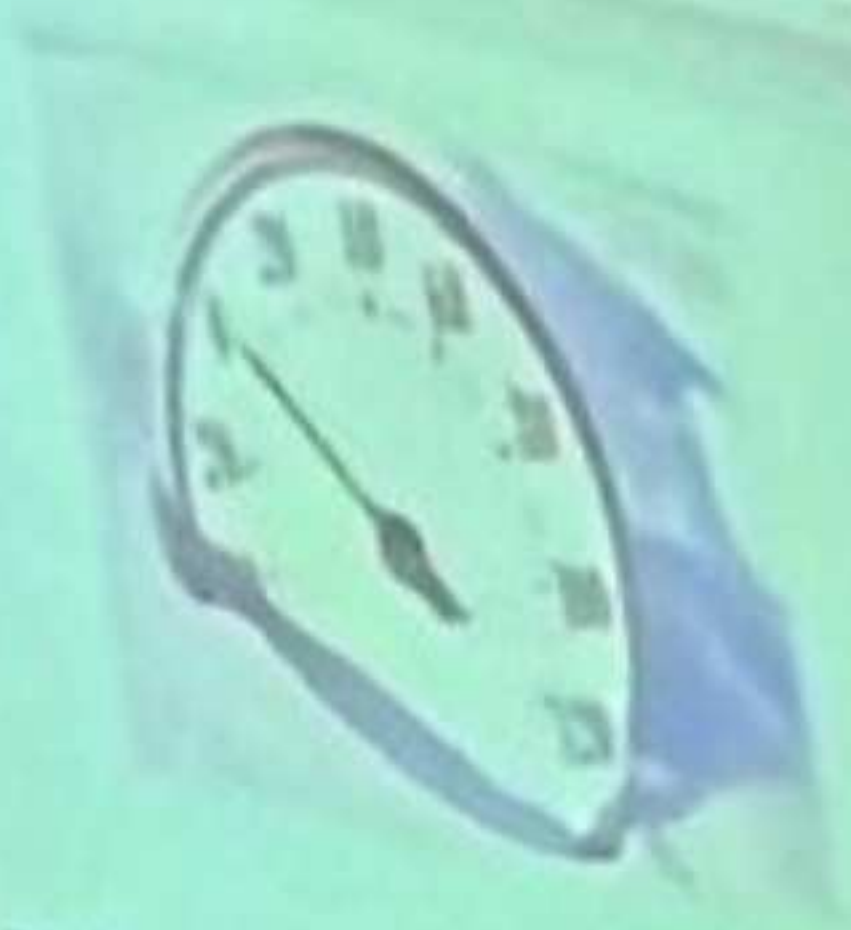
Interval Variables

Interval Variables:

- These have meaningful differences between values but no true zero point.

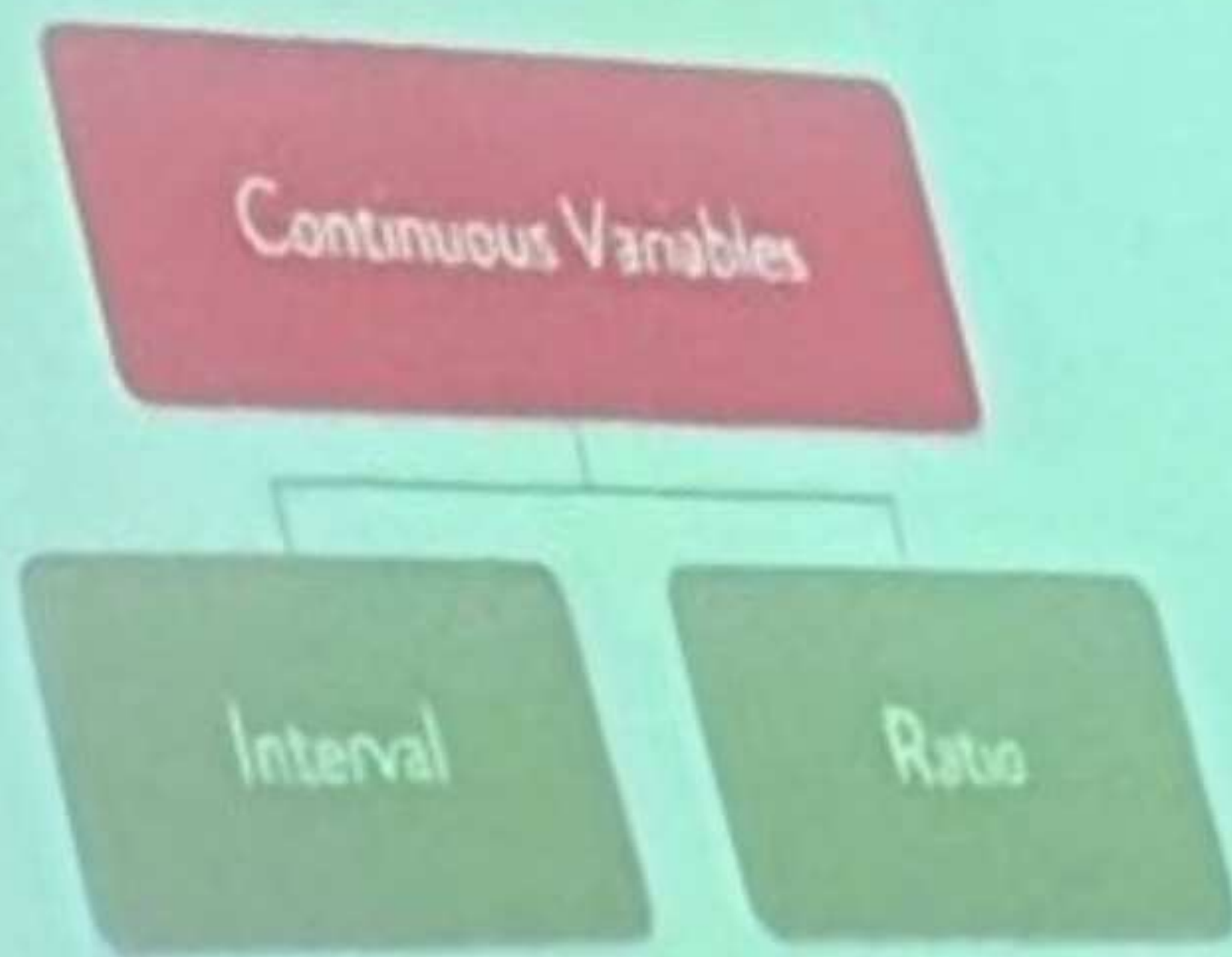
Example:

- Body temperature in Celsius (37°C , 39°C).
- Calendar years (0 is the birth of Christ) – 2024
- Time of day.



Continuous: Interval and Ratio Variables

- Both are types of continuous variables but have different characteristics:



Think?

- Is 20°C twice as hot as 10°C ?



- $20^{\circ}\text{C} = 68^{\circ}\text{F}$

- $10^{\circ}\text{C} = 50^{\circ}\text{F}$

- 20°C is not twice as hot as 10°C ; it simply represents a higher point on the temperature scale.

Ratio Variables

Ratio Variables:

- These have a true zero point, allowing for a comparison of magnitudes.

Example

- Age of a cat in years (e.g., 0 years means no age, and you can say a 5-year-old cat is twice as old as a 2.5-year-old cat).
- Concentration of a drug in the bloodstream
- Heart rate.
- Farm income.



ID, Name

1 = Engie

2 = Fatema

3 = Mariam

4 = Sama

Unknown quantitative difference

Independent and Dependent Variables

According to their role in statistical analysis

Dependent Variables

Outcome variable

Blood glucose level

Independent Variables

Predictor variables

Diet, exercise, medication

Identify the
type of variable

- Nominal
- Ordinal
- Discrete
- Continuous

ABO blood groups.

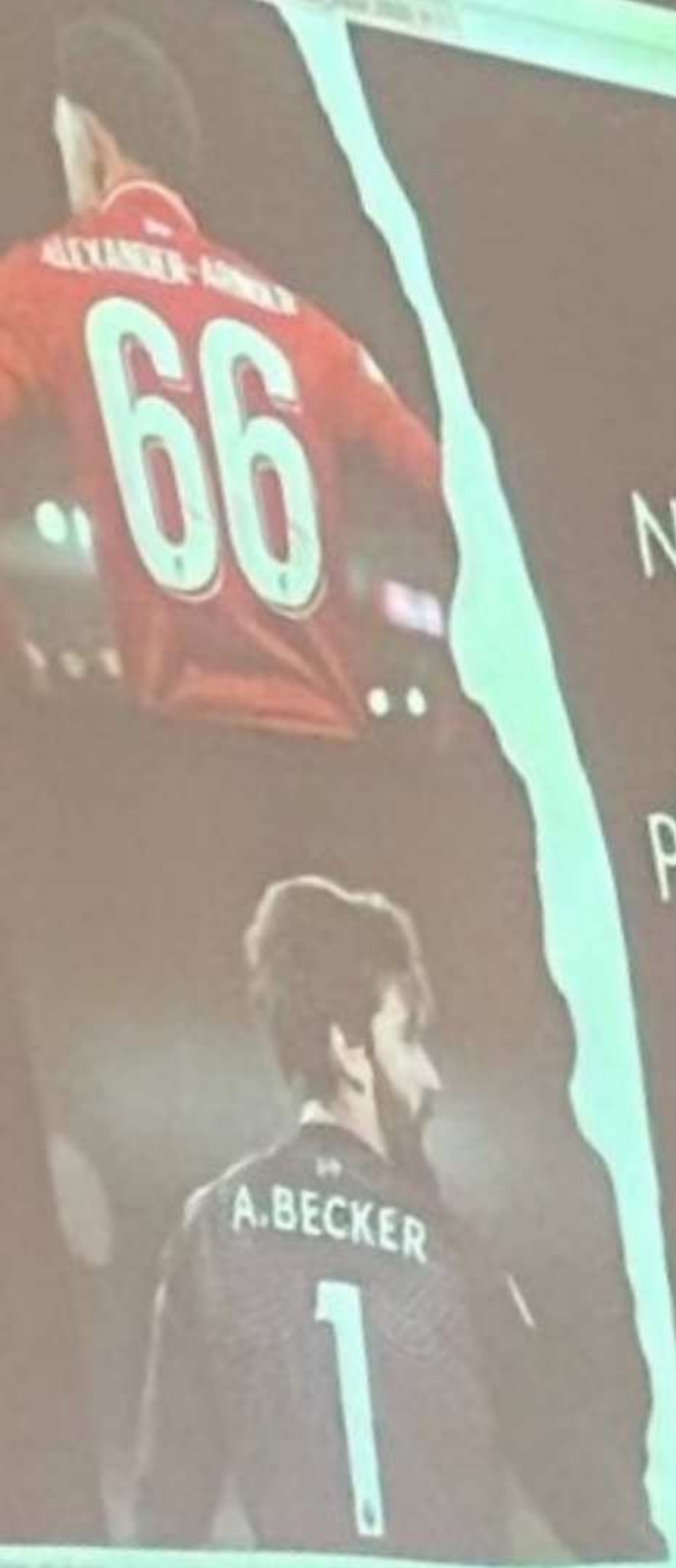
Egg production per week.

Body condition score.

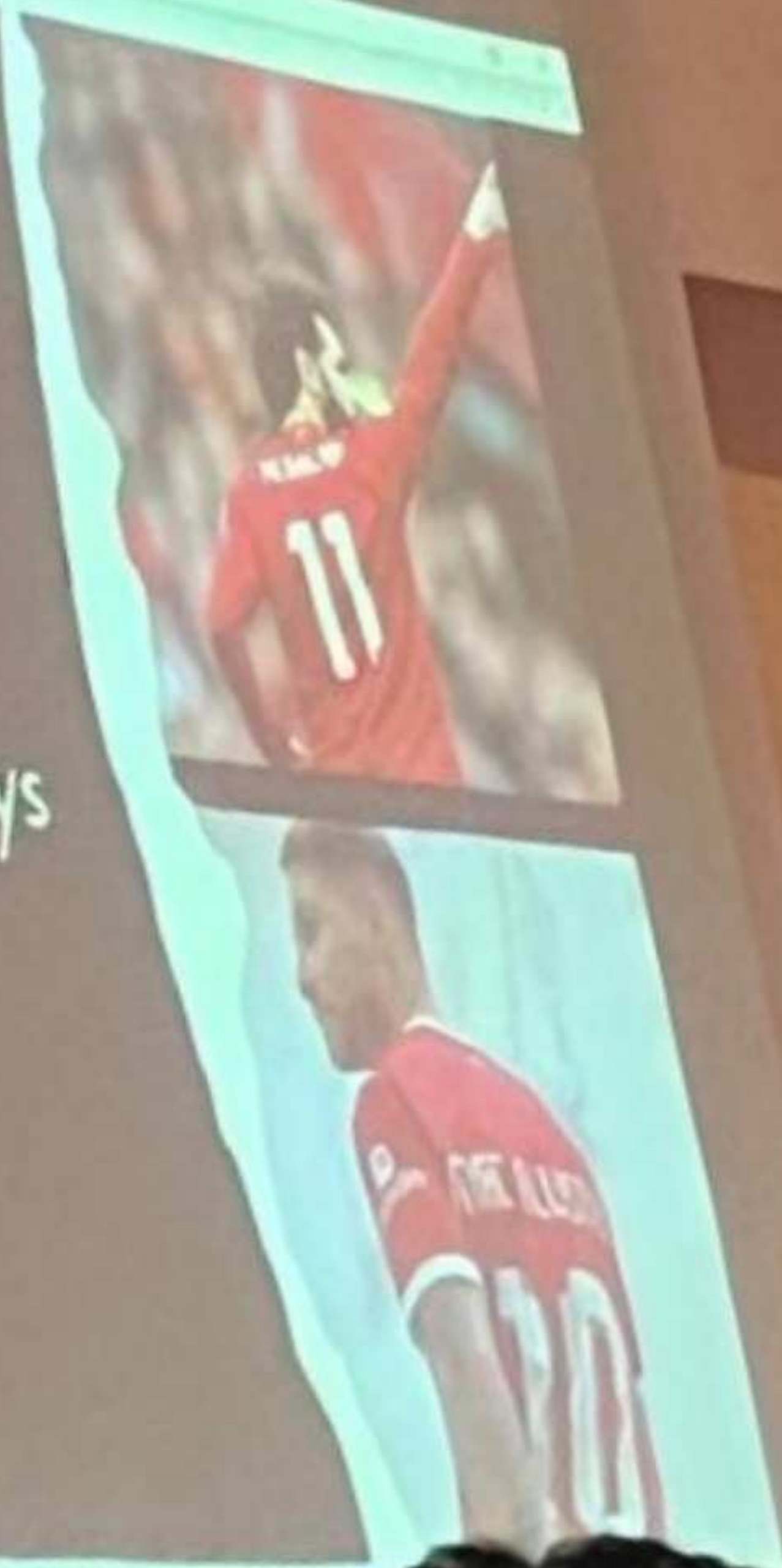
Plasma insulin level.

Yes or No.

Services per conception.



Numbers on Liverpool players' jerseys



Important Announcement

- Join Biostatistics and Computer on Microsoft Teams.
- Data files for SPSS will be sent.
- A quiz will be soon available this week.



Likert Scale



I am an angry person

1 = Strongly disagree

2 = disagree

3 = Neutral

4 = Agree

5 = Strongly agree

Unknown quantitative difference

Numerical Summaries of Data

Descriptive statistics

Measures of center

- Mean
- Median
- Mode

Measures of variation (dispersion)

- Range
- Variance
- Standard deviation
- Coefficient of variation
- IQR

Q	Gender	Height	Weight	Age	Category	Score	Label	Count
1	M	1.72	75	25	Normal	1	M	1
2	F	1.65	60	20	Normal	1	F	1
3	M	1.78	85	30	Obese	2	M	2
4	F	1.60	55	18	Normal	1	F	1
5	M	1.75	70	22	Normal	1	M	1
6	F	1.68	65	24	Normal	1	F	1
7	M	1.70	72	26	Normal	1	M	1
8	M	1.73	78	27	Obese	2	M	2
9	F	1.62	58	19	Normal	1	F	1
10	M	1.76	80	28	Obese	2	M	2

EXAMPLE

Methods for summarizing and presenting data

- (a) Numbers
- (b) Tables and graphs.

The arithmetic mean

The following are milk yield (kg) per day of five goats

1.5	4.5	4.0	5.3	3.5	4.0	6.1
-----	-----	-----	-----	-----	-----	-----

$$\bar{X} = \frac{\sum X_i}{n}$$

$$\bar{X} = \frac{1.5 + 4.5 + 4.0 + 5.3 + 3.5 + 4.0 + 6.1}{7} = 4.13 \text{ kg}$$

$$\sum (X_i - \bar{X}) = 0$$

The Weighted Mean

- Weighted mean is used when some scores contribute more "weight" than others in the final mean.

$$M_w = \frac{\sum(w \cdot x)}{\sum w}$$

- Example: CGPA calculation

If n is an **odd** number, the median is the value of the observation:

$$\left(\frac{n+1}{2} \right)^{\text{th}}$$

If n is an **even** number, the median is the arithmetic mean of the observations:

$$\left(\frac{n}{2} \right)^{\text{th}} \text{ and } \left(\frac{n}{2} + 1 \right)^{\text{th}}$$

The median

- The median is the value of the observation that is in the middle when observations are sorted from smallest to largest.

The Weighted Mean

Subject	Genetics	Biostatistics	Physiology	Anatomy	Biochemistry	Immunology	English	Behavior
GPA	2.666	4.000	2.333	2.000	3.556	3.000	4.000	2.666
CH	3	2	2	3	2	3	1	2

CGPA

$$= \frac{(2.666 \times 3) + (4.000 \times 2) + (2.333 \times 2) + (2.000 \times 3) + (3.556 \times 2) + (3.000 \times 3) + (4.000 \times 1) + (2.666 \times 2)}{3 + 2 + 2 + 3 + 2 + 3 + 1 + 2}$$

CGPA = 2.907

Sorting data from the smallest to the largest

1.5	3.5	4.0	4.0	4.5	5.3	6.1
-----	-----	-----	-----	-----	-----	-----

$n = 7$ (odd number), the median is the value of the observation

$$\left(\frac{n+1}{2}\right)^{th} = \left(\frac{7+1}{2}\right)^{th} = 4^{th} = 4$$

- Consider that the data set has 8 values.

1.5	4.5	4.0	5.3	3.5	4.0	6.1	4.8
-----	-----	-----	-----	-----	-----	-----	-----

- Arrange the data from the smallest to the largest.

1.5	3.5	4.0	4.0	4.5	4.8	5.3	6.1
-----	-----	-----	-----	-----	-----	-----	-----

- Since $n = 8$ (even number), the median is the average of

Mean or Median?

1. The median better indicates the center if the data has outlier(s).
(An outlier is a value that differs greatly from other values.)

1.5	3.5	4.0	4.0	4.5	4.8	5.3	6.1	25
-----	-----	-----	-----	-----	-----	-----	-----	----

2. With ordinal data (e.g., hemorrhage score)

- $\left(\frac{n}{2}\right)^{\text{th}}$ and $\left(\frac{n}{2} + 1\right)^{\text{th}}$

- $\left(\frac{8}{2}\right)^{\text{th}}$ and $\left(\frac{8}{2} + 1\right)^{\text{th}} = 4^{\text{th}}$ and 5^{th}

1.5	3.5	4.0	4.0	4.5	4.8	5.3	6.1
-----	-----	-----	-----	-----	-----	-----	-----

- The median = $\left(\frac{4 + 4.5}{2}\right) = 4.25 \text{ kg}$

The mode

- The mode is the value with the highest frequency in a data set.

1.5	3.5	4.0	4.0	4.5	4.8	5.3	6.1
-----	-----	-----	-----	-----	-----	-----	-----

- The mode is 4 kg.

Measures of variation (dispersion)

- Consider the following three data sets

Set	Values							Minimum	Maximum
A	50	50	50	50	50	50		50	50
B	30	40	50	60	70	50		30	70
C	0	30	50	70	100	50		0	100

- To better understand the nature of the data, you need to know how values vary.

Sorting data from the smallest to the largest

1.5	3.5	4.0	4.0	4.5	5.3	6.1
-----	-----	-----	-----	-----	-----	-----

$n = 7$ (odd number), the median is the value of the observation

$$\left(\frac{n+1}{2} \right)^{th} = \left(\frac{7+1}{2} \right)^{th} = 4^{th} = 4$$

Variance (S^2, σ^2)

$$S^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

why squared?

$$S^2 = \frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n - 1}$$

$$S^2 = \frac{132.05 - \frac{(28.9)^2}{7}}{7 - 1} = \frac{12.73}{6} = 2.12 \text{ kg}^2$$

1.5	4.5	4.0	5.3	3.5	4.0	6.1
-----	-----	-----	-----	-----	-----	-----

The Range

The Range

Range = Largest - Smallest

The example of the milk yield of goats

1.5	4.5	4.0	5.3	3.5	4.0	6.1	4.8
-----	-----	-----	-----	-----	-----	-----	-----

Range = $6.1 - 1.5 = 4.6$ kg

Standard Deviation (S)

$$S = \sqrt{S^2}$$

$$S = \sqrt{2.12} = 1.46 \text{ kg}$$

Data:

The time (h) taken by 19 mice to show signs of illness after ingesting a toxic plant.

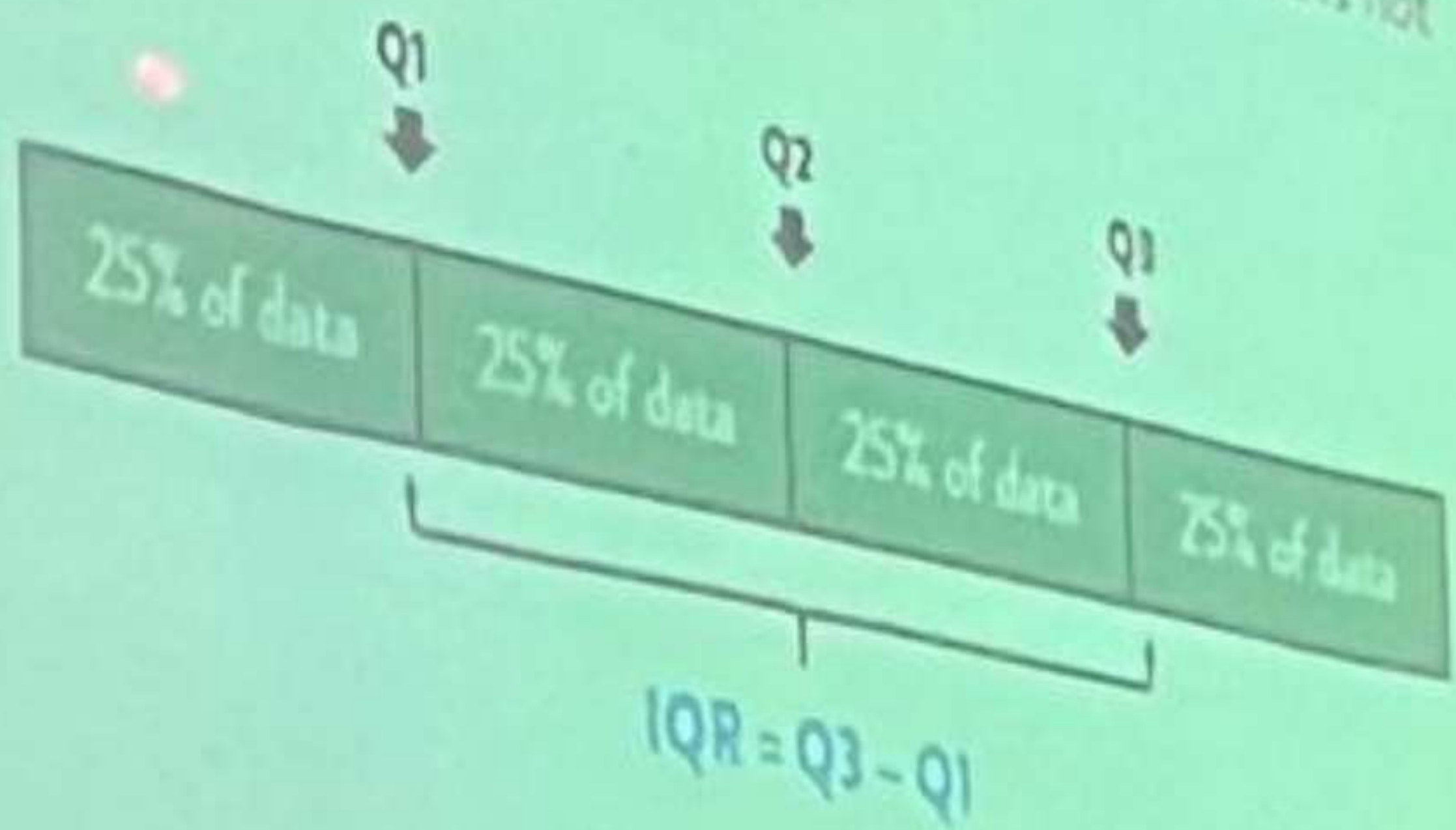
60	70	85	65	100	60	110	45	80	40	105	80	115	90	50	45	90	115	50
----	----	----	----	-----	----	-----	----	----	----	-----	----	-----	----	----	----	----	-----	----

Sort the data in increasing order.

the data in increasing order.																		
40	45	45	50	50	60	60	65	70	80	80	85	90	90	100	105	110	115	115
Q1					Q2					Q3								

The Interquartile Range (IQR)

- The interquartile range (IQR) is a measure of variation that is not sensitive to extreme values.



S or IQR

- Like the median, IQR is used for
 - Data with outlying values
 - Ordinal data
- In summary
 - Normally distributed data are best described with the mean and the standard deviation.
 - Median and IQR are preferred for non-normal and ordinal data.

Standard error of mean (SE)

- How close your sample mean to the population mean? One answer is the SE of the mean.
- SE measures the accuracy with which a sample represents a population.
- A small SE = the sample mean is an accurate reflection of the actual population mean.

$$SE = \frac{s}{\sqrt{n}}$$

Coefficient of Variation (CV)

The CV is a relative measure of variation.

For milk yield of goats

$$CV\% = \frac{S}{\bar{X}} \times 100$$

$$CV\% = \frac{1.46}{4.13} \times 100 = 35.4\%$$

For plasma glucose

$$CV\% = \frac{S}{\bar{X}} \times 100$$

$$CV\% = \frac{9}{45} \times 100 = 20\%$$

Data:

The time (h) taken by 19 mice to show signs of illness after ingesting a toxic plant

60	70	85	65	100	60	110	45	80	40	105	80	115	90	50	45	90	115	50
----	----	----	----	-----	----	-----	----	----	----	-----	----	-----	----	----	----	----	-----	----